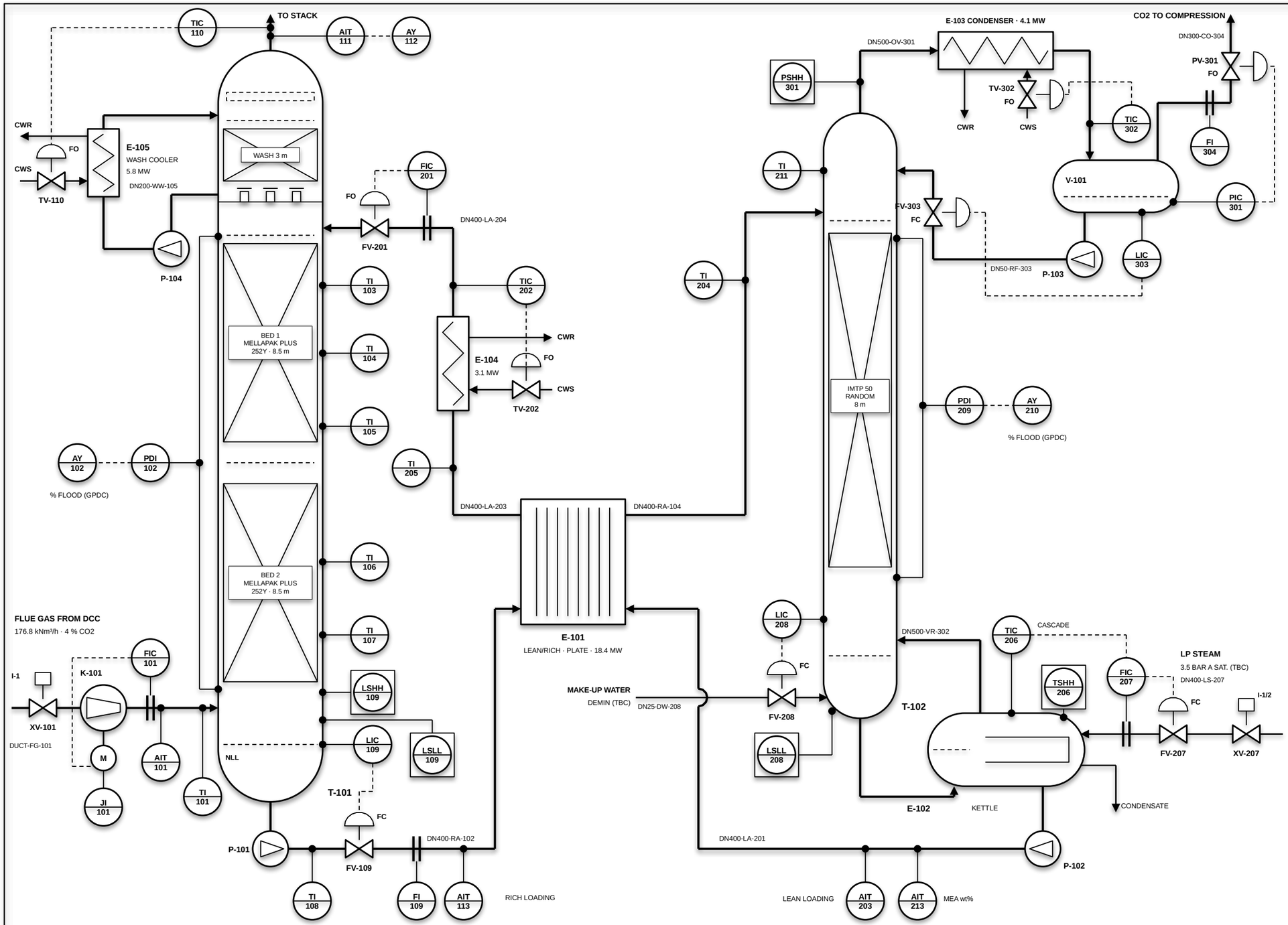



## LEGEND

- CONTROL VALVE · FC/FO = FAIL CLOSED/OPEN
- SHUTDOWN VALVE / DAMPER (SIS), SPRING-CLOSE
- FLOW ELEMENT (ORIFICE)
- CENTRIFUGAL PUMP
- CENTRIFUGAL BLOWER, VFD MOTOR (M)
- PACKED BED
- DCS INSTRUMENT (SHARED DISPLAY / CONTROL)
- SIS TRIP SWITCH (INDEPENDENT)
- INSTRUMENT SIGNAL
- PROCESS LINE (GAS, AMINE, CO2)
- UTILITY LINE (STEAM, CW, WATER)
- CONNECTION (TEE)
- CROSSING, NOT CONNECTED

**LINE NUMBER: DNxxx-SVC-nnn (SIZES TBC)**

FG FLUE GAS · RA RICH AMINE · LA LEAN AMINE · WW WASH WATER
OV OVERHEAD · VR REBOILER VAPOUR · RF REFLUX · CO CO2
LS LP STEAM · DW DEMIN WATER · CW COOLING WATER

**EQUIPMENT LIST (OTS DESIGN POINT, NOTE 3)**

| | |
|---|---|
| **K-101** | FLUE GAS BLOWER, VFD · 176.8 kNm³/h · 313 kW |
| **T-101** | ABSORBER · D 5.4 m · 2 × 8.5 m MELLAPAK PLUS 252Y<br>+ 3 m WATER WASH · 62 % FLOOD · ΔP 4.0 kPa |
| **E-105** | WASH WATER COOLER · 5.8 MW · CW |
| **P-104** | WASH WATER PUMP |
| **P-101** | RICH AMINE PUMP · 282 t/h |
| **E-101** | LEAN/RICH EXCHANGER, PLATE · 18.4 MW · ΔT 11 K |
| **E-104** | LEAN AMINE COOLER · 3.1 MW · CW |
| **T-102** | STRIPPER · D 2.6 m · 8 m IMTP 50 · 51 % FLOOD |
| **E-102** | KETTLE REBOILER · 13.7 MW · LP STEAM 22.9 t/h |
| **E-103** | OVERHEAD CONDENSER · 4.1 MW · CW |
| **V-101** | REFLUX DRUM · 175 kPa a · 40 °C |
| **P-103** | REFLUX PUMP · 5.6 t/h |
| **P-102** | LEAN AMINE PUMP · 270 t/h |

**DESIGN POINT (OTS MODEL)**

CAPTURE 90.0 % · CO2 12.5 t/h (300 t/d) · SRD 3.94 GJ/t
LEAN 269.7 t/h, α 0.246 · RICH α 0.471 · MEA 30 wt%
T-101 BULGE 57 °C · T-102 120 °C / 175 kPa a

**GENERAL NOTES**

1. DESIGN BASIS: 300 t/d CO2 AT 90 % CAPTURE FROM A SLIP-STREAM OF NGCC FLUE GAS AFTER THE DCC: 176.8 kNm³/h, 4.0 mol% CO2 (RANGE 3–6 %), 40 °C, SATURATED.
2. SOLVENT 30 wt% MEA (CO2-FREE BASIS). LOADINGS LEAN 0.246 / RICH 0.471 mol CO2/mol MEA; LEAN 269.7 t/h AT 40 °C; STRIPPER 175 kPa a, REBOILER 120 °C.
3. ALL NUMBERS COME FROM THE CALIBRATED OTS MODEL (MEA CO2 CAPTURE OTS VER 1). THEY ARE TRAINING VALUES, NOT A VENDOR DESIGN.
4. ASSUMED (TBC): LP STEAM 3.5 BAR A SATURATED; CW 30 °C; DEMIN MAKE-UP; E-101 PLATE TYPE; STRIPPER HETP 1.6 m; ALL LINE SIZES.
5. HOLD: PSVs (T-102, V-101, E-102 SHELL, E-101), VESSEL DESIGN PRESSURES AND RELIEF CASES ARE NOT SHOWN.
6. MATERIALS (TBC): SS 316L FOR HOT OR LOADED AMINE (T-102, E-101, E-102, V-101, PACKING); CS + CA FOR COLD LEAN SERVICE.
7. INSTRUMENTS C/W ROOT & TEST VALVES. DRAINS, VENTS, SAMPLE POINTS, CARBON FILTER, RECLAIMER, ANTIFOAM AND AMINE STORAGE NOT SHOWN.
8. DCS CALCULATIONS: AY-102 / AY-210 % FLOOD (GPDC); AY-112 CAPTURE FROM AIT-101/111; AY-214 CAPTURE FROM FI-304; AY-212 SRD.
9. COLUMN HEIGHTS NOT TO SCALE. T-101: 17 m PACKED + 3 m WASH; T-102: 8 m PACKED.

**CONTROL AND INTERLOCKS**

I-1 UNIT TRIP (LATCHED): LSHH-109 (90 %) OR ESD → K-101 STOP, XV-101 DAMPER SHUT, XV-207 STEAM SHUT. SOLVENT KEEPS CIRCULATING.
I-2 REBOILER TRIP: PSHH-301 (250 kPa a), TSHH-206 (130 °C) OR LSLL-208 (10 %) → XV-207 SHUT. RESET WHEN CLEAR.
I-3 PUMP PROTECTION: LSLL-109 (5 %) STOPS P-101; LSLL-208 (10 %) STOPS P-102.
A. FIC-101 SETS THE FLUE GAS BY K-101 SPEED; FIC-201 SETS L/G; TIC-202 HOLDS THE LEAN AT 40 °C.
B. TIC-206 (120 °C) CASCADES TO FIC-207 (STEAM); PIC-301 (175 kPa a) THROTTLES PV-301 ON THE CO2 PRODUCT.
C. LIC-109 → FV-109 RICH; LIC-208 → FV-208 MAKE-UP; LIC-303 → FV-303 REFLUX; TIC-110 → TV-110 WASH (WATER BALANCE).
D. K-101 STARTS ONLY WITH THE SIS IN RUN AND FIC-201 ≥ 120 t/h. STRIP TO AIT-203 ≤ 0.26 BEFORE ADMITTING GAS.
E. LIMITS: REBOILER ≤ 125 °C (MEA DEGRADATION); FLOOD < 80 %; MEA 28–32 wt% (AIT-213).
F. FAIL POSITIONS: FV-201, TV-110, TV-202, TV-302, PV-301 FO; FV-109, FV-207, FV-208, FV-303 FC.

| REV | DESCRIPTION | DATE |
|---|---|---|
| A | ISSUED FOR OTS (TRAINING) | 25-SEP-26 |

**CHEM ENG CHULALONGKORN UNIVERSITY · OTS**

**MEA CO2 CAPTURE · ABSORBER / STRIPPER**

| DRAWN: A.S. | CHK'D: -- | APP'D: -- | DATE: 25-SEP-26 |
|---|---|---|---|
| **DWG NO. CC-PID-101** | REV A | SCALE N.T.S. | SHEET 1 OF 1 |